The first asymmetric Sonogashira coupling for the enantioselective generation of planar chirality in paracyclophanes†

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The double Sonogashira coupling of diiodoparacyclophanes with alkynes proceeded to give planarly chiral dialkynylparacyclophanes; a chiral Pd catalyst, which was prepared *in situ* from $PdCl_2(CH_3CN)_2$ and Taniaphos, realized the first asymmetric Sonogashira coupling with up to *ca*. 80% ee.

Pd-catalyzed cross couplings are some of the most conventional and reliable carbon–carbon bond forming reactions,[1] and they have made a significant contribution to the synthesis of complex molecules with various functionalities.[2] They can be categorized into patterns of couplings, such as C_{sp^2}–C_{sp^2}, C_{sp^2}–C_{sp^3}, C_{sp}–C_{sp^2}, and so on. The asymmetric versions, which use chiral Pd catalysts, have been a challenging topic in asymmetric synthesis.[3] The pioneering work in this area was the dynamic kinetic resolution of the reaction between secondary alkyl metal reagents and vinyl or aryl halides; the coupling of a Grignard reagent with a vinyl bromide (C_{sp^3}–C_{sp^2} coupling) generated an asymmetric carbon center.[4] Enantioselective desymmetrization is a different approach;[5] the coupling of aryl or vinyl metal reagents with a *meso*-dichloroarene–Cr complex (C_{sp^2}–C_{sp^2} coupling) provided planarly chiral products with a moderate enantioselectivity.[5a–c] In the coupling of aryl Grignard reagents with a *meso*-biaryl ditriflate (C_{sp^2}–C_{sp^2} coupling), axially chiral biaryl compounds were obtained in a highly enantioselective manner.[5d,e] The enantioselective coupling of aryl metal reagents with aryl halides (C_{sp^2}–C_{sp^2} coupling), which induces axial chirality, is another successful example.[6] The asymmetric Buchwald–Hartwig amination has also been reported, in which the coupling of amides with aryl halides (N_{sp^3}–C_{sp^2} coupling) gave axially chiral anilides with a high ee.[7] Recently, Ni-catalyzed enantioselective couplings of alkyl metal reagents with secondary electrophiles, such as allylic halides, benzyl halides and α-bromoesters (C_{sp^3}–C_{sp^3} coupling), have also been reported.[8] However, it is difficult to induce a new chiral motif in C_{sp}–C_{sp^2} couplings because of the rigid and linear alkynyl moiety,[9] and asymmetric Sonogashira couplings have never previously been reported.

This Communication demonstrates the first example of an asymmetric Sonogashira coupling (C_{sp}–C_{sp^2} coupling) in the enantioselective synthesis of chiral paracyclophanes. Our concept for the asymmetric induction of planar chirality is shown in Fig. 1; when the *ansa* chain of a dihaloparacyclophane is sufficiently long, interconversion between its enantiomers readily occurs, even at room temperature. In the doubly-alkynylated product, however, planar chirality must be generated, because the rigid and linear alkynyl moieties serve as "props", which prevent interconversion from occurring. This can be categorized as a new type of dynamic kinetic resolution.[10]

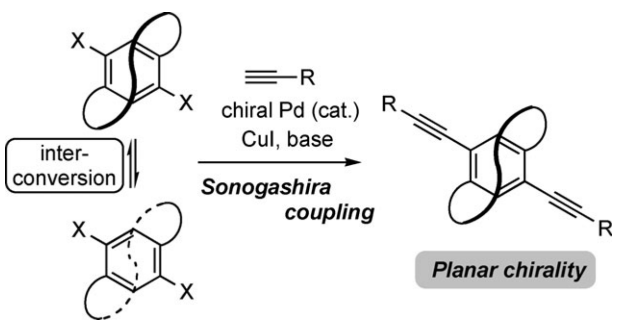

Fig. 1 Our concept of asymmetric Sonogashira coupling.

We synthesized [6.6]paracyclophane **1a**, and the ^{1}H-NMR spectrum of its benzylic methylene protons (dashed circles in Fig. 2) gave a multiplet (left spectrum). This suggests that the two enantiomers of the cyclophane do not freely interconvert, and they were observed by HPLC analysis using a chiral column.[11] In contrast, the ^{1}H-NMR spectrum of the corresponding protons of [7.7]paracyclophane **1b** gave a broad signal (Fig. 2, right spectrum) and, HPLC analysis under the same conditions showed a single peak.[12] Based on these results, we assumed that the enantiomers of paracyclophane **1b** can freely interconvert at room temperature and submitted it to asymmetric Sonogashira coupling.

We chose *p*-ethynylanisole as a model alkyne and examined its double coupling with **1b** under the reaction conditions of Sonogashira coupling ($PdCl_2(CH_3CN)_2$, CuI and Cs_2CO_3 as a base[13]), using (*R*)-BINAP as a chiral ligand. The doubly-alkynylated product, **2ba**, was obtained in an excellent yield with a moderate ee of 29% (Table 1, entry 1).[14] When (*S*)-BINAP was used, the opposite enantiomer was the major

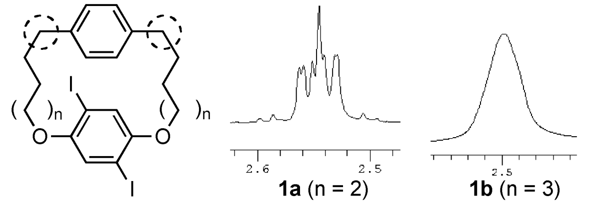

Fig. 2 The ^{1}H-NMR spectra of paracyclophanes **1a** and **1b**.

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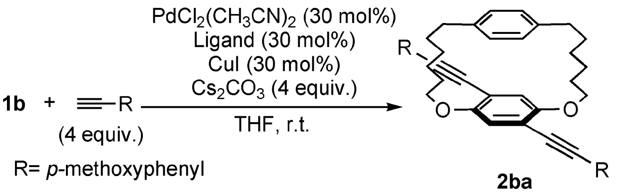
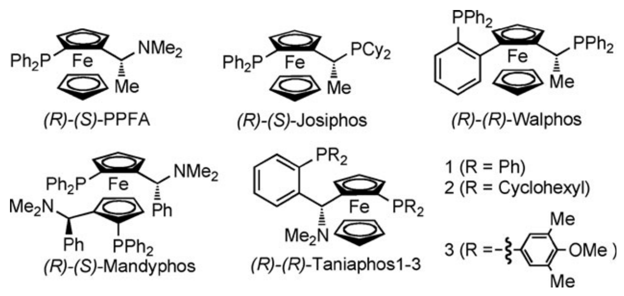
Table 1 Screening of chiral ligands for the asymmetric Sonogashira coupling of cyclophane **1b**



Entry	Ligand	Time/h	Yield (%)	ee (%)
1	(*R*)-BINAP	18	99	29
2	(*S*)-BINAP	15	95	−35
3	(*S*)-tol-BINAP	27	98	5
4	(*S*)-H_8-BINAP	9	98	−14
5	(*S*)-SEGPHOS[a]	10	88	−4
6	(*R*)-(*S*)-PPFA	6	93	<2
7	(*R*)-(*S*)-Josiphos	24	33	−7
8	(*R*)-(*R*)-Walphos	48	33	6
9	(*R*)-(*S*)-Mandyphos	6	85	−26
10	**(*R*)-(*R*)-Taniaphos 1**	**6**	**98**	**57**
11	(*R*)-(*R*)-Taniaphos 2	18	66	34
12	(*R*)-(*R*)-Taniaphos 3	6	91	44

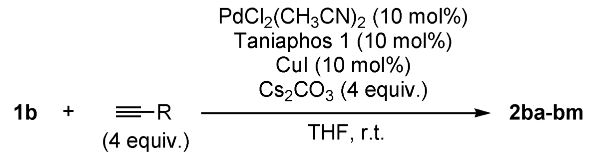
[a] SEGPHOS: 5,5′-bis(diphenylphosphino)-4,4′-bi-1,3-benzodioxole.



product (Table 1, entry 2), which meant that our concept of the enantioselective generation of planar chirality had been realized. BINAP derivatives also achieved high to excellent yields, but the enantioselectivities were low (Table 1, entries 3–5). In the case of PPFA, the enantioselectivity was low, but the reaction was concluded within a shorter time (Table 1, entry 6). Thus, we examined other ferrocene-based ligands: Josiphos and Walphos gave poor results in terms of both yield and enantioselectivity (Table 1, entries 7 and 8), and Mandyphos gave a good yield, although the enantioselectivity was not improved (Table 1, entry 9). Taniaphos 1 achieved the best enantioselectivity, with an excellent yield (Table 1, entry 10). After screening Taniaphos derivatives (Table 1, entries 11 and 12), we chose Taniaphos 1 for further investigations.

We used 10 mol% of chiral Pd catalyst and screened various alkynes in the asymmetric Sonogashira couplings with **1b** (Table 2). The coupling with phenylacetylene gave dialkynylated product **2bb** in a 56% ee (Table 2, entry 1). In the case of *o*-ethynylanisole, a higher enantioselectivity was achieved than with *m*- or *p*-ethynylanisole (Table 2, entries 2–4). The coupling of **1b** with various *ortho*-substituted ethynylbenzenes was examined (Table 2, entries 5–9), and the enantioselectivities were more than 70% when 1-bromo-2-ethynylbenzene and 1-acetyl-2-ethynylbenzene were used (Table 2,

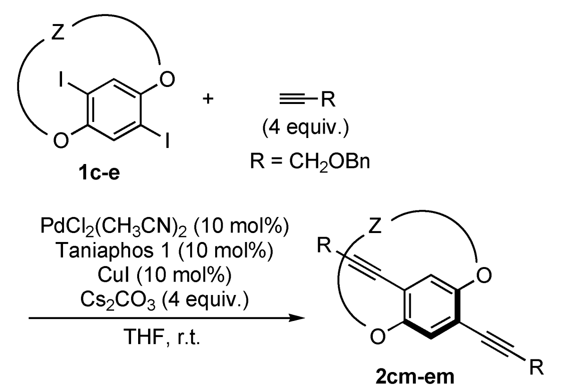
Table 2 The asymmetric Sonogashira coupling of cyclophane **1b** with various alkynes

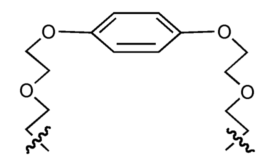
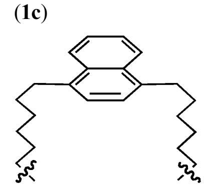


Entry	R	Time/h	Yield (%)	ee (%)
1	Phenyl	18	87 (**2bb**)	56
2	*p*-Methoxyphenyl	12	91 (**2ba**)	52
3	*m*-Methoxyphenyl	24	92 (**2bc**)	59
4	*o*-Methoxyphenyl	48	92 (**2bd**)	69
5[a]	*o*-Methylphenyl	18	90 (**2be**)	62
6	2-Biphenyl	51	90 (**2bf**)	61
7	*o*-Chlorophenyl	72	81 (**2bg**)	56
8	*o*-Bromophenyl	72	85 (**2bh**)	72
9	*o*-Acetylphenyl	40	71 (**2bi**)	77
10	2,6-Dimethoxyphenyl	18	99 (**2bj**)	66
11	2,6-Dibromophenyl	72	71 (**2bk**)	72
12	CH_2OMe	72	73 (**2bl**)	79
13	CH_2OBn	96	87 (**2bm**)	78

[a] 6 equiv. of alkyne were used.

Table 3 The asymmetric Sonogashira coupling of cyclophanes with different *ansa* chains



Entry	Z	Time/h	Yield (%)	ee (%)
1	 (**1c**)	48	81 (**2cm**)	78
2	 (**1d**)	54	91 (**2dm**)	73
3	$-(CH_2)_{16}-$ (**1e**)	48	89 (**2em**)	77

entries 8 and 9). The enantioselectivity of a 2,6-disubsutituted ethynylbenzene was comparable to that of a 2-monosubstituted example (Table 2, entries 10 and 11). Furthermore, the coupling of **1b** with propargyl ethers achieved the best enantioselectivity of *ca.* 80% (Table 2, entries 12 and 13).

Next, we examined the coupling of other paracyclophanes with benzyl propargyl ether (Table 3). Hexaoxa[7.7]paracyclophane **1c**, which has six oxygen atoms on its *ansa* chain (Table 3, entry 1), [7](1,4)naphthaleno[7]paracyclophane **1d** (Table 3, entry 2) and [18]paracyclophane **1e** with an aliphatic *ansa* chain (Table 3, entry 3), were examined. The corresponding dialkynylated products were obtained in good to high yields, with enantioselectivities as good as those of **1b**.

In conclusion, we have developed an asymmetric Sonogashira coupling of diiodoparacyclophanes to give planarly chiral dialkynylparacyclophanes with *ca.* 80% ee. The present results are also notable from a synthetic point of view, because planarly chiral paracyclophanes are useful chiral motifs and functional materials;[15] PHANEPHOS derivatives as chiral ligands[16] and parapyridinophane derivatives as co-enzyme models[17] are the selected examples. However, there has been only one report of enantioselective and catalytic syntheses of chiral paracyclophanes.[18] Therefore, we have demonstrated the enantioselective synthesis of chiral paracyclophanes based on the new concept of asymmetric induction, and its use in other reactions is under investigation.

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Notes and references

1 (*a*) *Handbook of Organopalladium Chemistry for Organic Synthesis*, ed. E. Negishi and A. De Meijere, Wiley-Interscience, New York, USA, 2002; (*b*) J. Tsuji, *Palladium Reagents and Catalysts: New Perspectives for the 21st Century*, Wiley, Chichester, UK, 2004; (*c*) J. Tsuji, *Palladium in Organic Synthesis*, Springer, Berlin, Germany, 2005.

2 K. C. Nicolaou, P. G. Bulger and D. Sarlah, *Angew. Chem., Int. Ed.*, 2005, **44**, 4442.

3 For reviews, see: (*a*) T. Hayashi, in *Comprehensive Asymmetric Catalysis*, ed. E. N. Jacobsen, A. Pfaltz and H. Yamamoto, Springer, Berlin, Germany, 1999, vol. 2, pp. 888; (*b*) M. Ogasawara and T. Hayashi, in *Catalytic Asymmetric Synthesis*, ed. I. Ojima, Wiley-VCH, New York, USA, 2000, pp. 651; (*c*) T. Hayashi, *J. Organomet. Chem.*, 2002, **653**, 41.

4 (*a*) G. Consiglio and C. Botteghi, *Helv. Chim. Acta*, 1973, **56**, 460; (*b*) Y. Kiso, K. Tamao, N. Miyake, K. Yamamoto and M. Kumada, *Tetrahedron Lett.*, 1974, **15**, 3; (*c*) T. Hayashi, M. Tajika, K. Tamao and M. Kumada, *J. Am. Chem. Soc.*, 1976, **98**, 3718; (*d*) T. Hayashi, M. Fukushima, M. Konishi and M. Kumada, *Tetrahedron Lett.*, 1980, **21**, 79; (*e*) T. Hayashi, M. Konishi, M. Fukushima, T. Mise, M. Kagotani, M. Tajika and M. Kumada, *J. Am. Chem. Soc.*, 1982, **104**, 180; (*f*) T. Hayashi, M. Konishi, H. Ito and M. Kumada, *J. Am. Chem. Soc.*, 1982, **104**, 4962; (*g*) T. Hayashi, M. Konishi, M. Fukushima, K. Kanehira, T. Hioki and M. Kumada, *J. Org. Chem.*, 1983, **48**, 2195; (*h*) T. Hayashi, T. Hagihara, Y. Katsuro and M. Kumada, *Bull. Chem. Soc. Jpn.*, 1983, **56**, 363; (*i*) T. Hayashi, M. Konishi, Y. Okamoto, K. Kabeta and M. Kumada, *J. Org. Chem.*, 1986, **51**, 3772; (*j*) T. Hayashi, A. Yamamoto, M. Hojo and Y. Ito, *J. Chem. Soc., Chem. Commun.*, 1989, 495.

5 (*a*) M. Uemura, H. Nishimura and T. Hayashi, *Tetrahedron Lett.*, 1993, **34**, 107; (*b*) M. Uemura, H. Nishimura and T. Hayashi, *J. Organomet. Chem.*, 1994, **473**, 129; (*c*) K. Kamikawa, K. Harada and M. Uemura, *Tetrahedron: Asymmetry*, 2005, **16**, 1419; (*d*) T. Hayashi, S. Niizuma, T. Kamikawa, N. Suzuki and Y. Uozumi, *J. Am. Chem. Soc.*, 1995, **117**, 9101; (*e*) T. Kamikawa, Y. Uozumi and T. Hayashi, *Tetrahedron Lett.*, 1996, **37**, 3161.

6 For asymmetric Kumada couplings, see: (*a*) K. Tamao, A. Minato, N. Miyake, T. Matsuda, Y. Kiso and M. Kumada, *Chem. Lett.*, 1975, 133; (*b*) K. Tamao, H. Yamamoto, H. Matsumoto, N. Miyake, T. Hayashi and M. Kumada, *Tetrahedron Lett.*, 1977, **18**, 1389; (*c*) T. Hayashi, K. Hayashizaki, T. Kiyoi and Y. Ito, *J. Am. Chem. Soc.*, 1988, **110**, 8153; (*d*) T. Hayashi, K. Hayashizaki and Y. Ito, *Tetrahedron Lett.*, 1989, **30**, 215. For asymmetric Suzuki couplings, see: (*e*) A. N. Cammidge and K. V. L. Crépy, *Chem. Commun.*, 2000, 1723; (*f*) J. Yin and S. L. Buchwald, *J. Am. Chem. Soc.*, 2000, **122**, 12051; (*g*) A.-S. Castanet, F. Colobert, P.-E. Broutin and M. Obringer, *Tetrahedron: Asymmetry*, 2002, **13**, 659; (*h*) A. Herrbach, A. Marinetti, O. Baudoin, D. Guénard and F. Guéritte, *J. Org. Chem.*, 2003, **68**, 4897; (*i*) A. N. Cammidge and K. V. L. Crépy, *Tetrahedron*, 2004, **60**, 4377; (*j*) M. Genov, A. Almorín and P. Espinet, *Chem.–Eur. J.*, 2006, **12**, 9346. For asymmetric Negishi couplings, see: (*k*) M. Genov, B. Fuentes, P. Espinet and B. Pelaz, *Tetrahedron: Asymmetry*, 2006, **17**, 2593. For microwave-assisted asymmetric Suzuki and Negishi couplings, see: (*l*) M. Genov, A. Almorín and P. Espinet, *Tetrahedron: Asymmetry*, 2007, **18**, 625.

7 (*a*) O. Kitagawa, M. Kohriyama and T. Taguchi, *J. Org. Chem.*, 2002, **67**, 8682; (*b*) J. Terauchi and D. P. Curran, *Tetrahedron: Asymmetry*, 2003, **14**, 587; (*c*) O. Kitagawa, M. Takahashi, M. Yoshikawa and T. Taguchi, *J. Am. Chem. Soc.*, 2005, **127**, 3676; (*d*) O. Kitagawa, M. Yoshikawa, H. Tanabe, T. Morita, M. Takahashi, Y. Dobashi and T. Taguchi, *J. Am. Chem. Soc.*, 2006, **128**, 12923.

8 (*a*) C. Fischer and G. C. Fu, *J. Am. Chem. Soc.*, 2005, **127**, 4594; (*b*) F. O. Arp and G. C. Fu, *J. Am. Chem. Soc.*, 2005, **127**, 10482; (*c*) S. Son and G. C. Fu, *J. Am. Chem. Soc.*, 2008, **130**, 2756; (*d*) X. Dai, N. A. Strotman and G. C. Fu, *J. Am. Chem. Soc.*, 2008, **130**, 3302; (*e*) B. Saito and G. C. Fu, *J. Am. Chem. Soc.*, 2008, **130**, 6694. The enantioselective Ni-catalyzed cross coupling of secondary benzyl bromides with trialkynylindium (C_{sp^3}–C_{sp} coupling) has also been reported; see: (*f*) J. Caeiro, J. P. Sestelo and L. A. Sarandeses, *Chem.–Eur. J.*, 2008, **14**, 741.

9 Only one example of asymmetric C_{sp}–C_{sp^2} coupling has been reported: the enantioposition-selective coupling of a *meso*-biaryl ditriflate with an alkynyl Grignard reagent gave axially chiral biaryl compounds; see ref. 5*e*.

10 For reviews of dynamic kinetic resolution, see: (*a*) F. F. Huerta, A. B. E. Minidis and J.-E. Bäckvall, *Chem. Soc. Rev.*, 2001, **30**, 321; (*b*) H. Pellissier, *Tetrahedron*, 2003, **59**, 8291; (*c*) H. Pellissier, *Tetrahedron*, 2008, **64**, 1563.

11 Daicel Chiralpak IA: 4 × 250 mm, 254 nm UV detector, rt; eluent: 20% CH_2Cl_2 in hexane, flow rate: 1.0 mL min^{-1}.

12 As far as we investigated (Chiralpak IA, IB, AD, and Chiralcel OD, OJ), the HPLC analyses of compound **1b** showed a single peak.

13 The choice of base is very important: when triethylamine was used, the reaction proceeded sluggishly and the yield of dialkynylated product **2ba** was moderate (43%), along with the formation of monoalkynylated product (40%).

14 The ^{1}H-NMR spectrum of the benzylic methylene protons of **2ba** was a multiplet, and the enantiomers were observed by HPLC analysis using a chiral column (Daicel Chiralpak IA: 4 × 250 mm, 254 nm UV detector, rt; eluent: 30% CH_2Cl_2 in hexane, flow rate: 1.0 mL min^{-1}).

15 For reviews of cyclophanes, see: (*a*) *Cyclophane Chemistry*, ed. F. Vögtle, Wiley, Chichester, UK, 1993; (*b*) *Modern Cyclophane Chemistry*, ed. R. Gleither and H. Hopf, Wiley, Chichester, UK, 2004.

16 P. J. Pye, K. Rossen, R. A. Reamer, N. N. Tsou, R. P. Volante and P. J. Reider, *J. Am. Chem. Soc.*, 1997, **119**, 6207.

17 (*a*) N. Kanomata and T. Nakata, *Angew. Chem., Int. Ed. Engl.*, 1997, **36**, 1207; (*b*) N. Kanomata and T. Nakata, *J. Am. Chem. Soc.*, 2000, **122**, 4563.

18 A chiral Rh complex catalyzed the coupling of 1,4-bis(bromomethyl)benzene derivatives with dithiols to give chiral dithiaparacyclophanes (up to 60% ee): K. Tanaka, T. Hori, T. Osaka, K. Noguchi and M. Hirano, *Org. Lett.*, 2007, **9**, 4881.

